

The Making of a Radio-Controlled Plane



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If you are planning to make your own multi-rotor from scratch at home, note that choosing the right electronic speed control (ESC) for your radio-controlled electric aircraft (RC plane) is not an easy task. Although a number of interesting multi-rotor platforms, like monocopter, twincopter, tri-copter, quadcopter, hexacopter and octocopter, are available now, building one for fun flying/hovering completely depends on your imagination, skill and, of course, judicious selection and use of system components.

In any model aircraft, key components of the power system include an electric motor, electronic speed control and a battery pack. A perfectly-matched combination of the key components will bring out an efficient output from the power system. In fact, performance of the entire power system depends on the selection of the motor. Usually, brush-less motors (brush-less out-runner motors) are used in a power system.

Motor selection

There are a few vital considerations you should keep in mind when choosing the motor. The easiest choice is whether to use direct drive or a gear box, because if you are looking for high rpm (revolutions per minute), you will probably need a direct drive in-runner motor. For lower rpm, you can pick an out-runner in direct drive (or try an in-runner through a gearbox). In connection with this, always do your homework on other important things, like kV ratings ($\text{kV} = \text{rpm}/\text{V}$) of the motor, battery voltage and capacity of the battery pack, among

others. Some important specifications that come with brush-less motors are kV, max. current (A), shaft diameter, thrust, suggested propeller, lithium polymer (LiPo) (3S-4S) and weight.

With a known kV rating, you can calculate the rpm of the motor using the formula $\text{rpm} = \text{kV} \times \text{V}$, where V is battery voltage. The maximum current rating indication helps you pick the right ESC and battery pack. Similarly, the shaft diameter helps you select the right propeller. Thrust is the value of the force that the motor can produce with the suggested propeller. LiPo (3S-4S) indicates the motor's ability to run on 3-series (3S = 11.1V) and/or 4-series (4S = 14.8V) LiPo battery packs.

Rough calculation. If you are making a quadcopter for lifting 1000gm of payload, 2000gm thrust is a common requirement*. Since a quadcopter has four motors, each motor should handle at least 500gm of thrust to function better.

(*The basic formula you should keep in mind is that, AUW (all up weight) of a multi-rotor should be half of the total thrust, that is, total thrust (minimum) = $2 \times \text{AUW}$.)

Electronic speed control

For the ESC to work with your selected motor, it must be rated for voltage of your LiPo battery pack, and it must be able to handle the motor's voltage and current requirements. In principle, it is always better to have an ESC with a rating slightly higher (1.5x maximum) than the peak current (at full throttle) requirement of the motor.

All ESCs have inherent voltage and current limitations. So, select an ESC that is designed to work with an equal or higher voltage. It is not recommended to connect a high-voltage battery to a low-voltage ESC, and a high-voltage ESC to a low-voltage battery. Likewise, it is very important to know the peak current rate of the motor at full throttle. Always prefer an ESC with a current rating that is higher than the actual demand.

Brush-less out-runner motors

